

TAMIL NADU STATE BOARD - STANDARD 11 CHEMISTRY VOLUME 2

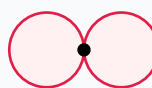
CHAPTER 13: HYDROCARBONS

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 11 English Medium
Subject:	Chemistry Volume 2
Chapter Name:	Hydrocarbons

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Atomic Orbitals Wave Functions (s & p)

p orbital (px)

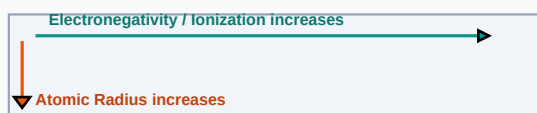
Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Wurtz Reaction	The coupling of two alkyl halides in the presence of sodium metal in dry ether to form higher alkanes.
Markownikoff Rule	In the addition of unsymmetrical reagents to unsymmetrical alkenes, the negative part adds to the carbon with fewer hydrogen atoms.
Aromaticity	The property of conjugated cyclic planar rings to show high stability due to $4n+2$ pi delocalized electrons (Huckel rule).

Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

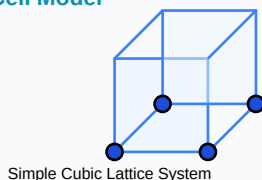
Periodic Properties Trends Direction Map



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:

Crystal Solid Lattice Unit Cell Model



Essential Formulas, Laws & Identities:

- Huckel's Rule: $4n + 2$ pi electrons = Aromatic
- Ozonolysis: Alkene + $O_3 \rightarrow$ Ozonide $\rightarrow$ Aldehydes / Ketones

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

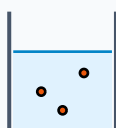
Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Convert Propyne to Propene.

Step-by-step Solution:

React Propyne ($\text{CH}_3\text{-C}\equiv\text{CH}$) with Lindlar catalyst (Pd/CaCO_3) under hydrogen gas to selectively reduce it to Propene ($\text{CH}_3\text{-CH=CH}_2$).

Titration Beaker & Solute Precipitation



Reacting Mixture

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

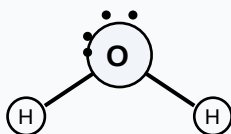
HOTS Challenge 1: Describe Electrophilic Substitution reactions in Benzene, giving mechanisms for Nitration and Friedel-Crafts Alkylation.

Analytical Solution Strategy:

(1) Nitration: Reagents $\text{HNO}_3/\text{H}_2\text{SO}_4$ generate electrophile NO_2^+ . NO_2^+ attacks benzene ring forming sigma complex, which loses H^+ to yield nitrobenzene.

(2) Alkylation: Reagents RCl/AlCl_3 generate R^+ . R^+ attacks ring to yield alkylbenzene.

Lewis Dot Structure (H_2O Bent Molecular Geometry)



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

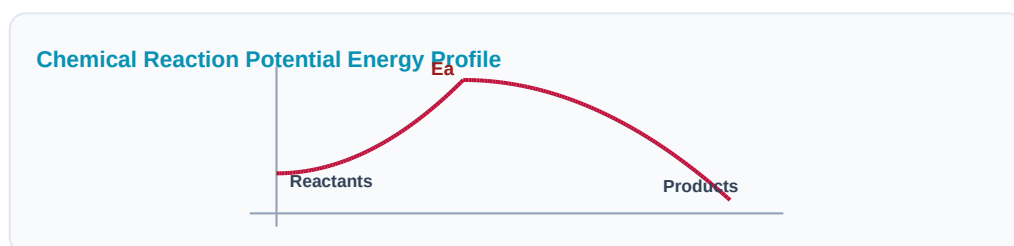
Q1 (1-Mark): Write the chemical equation for Kolbe's electrolytic synthesis.

Q2 (2-Mark): Predict the product: Propene + HBr in presence of peroxide (Kharasch effect).

Q3 (2-Mark): State Huckel's rule with examples of aromatic, non-aromatic, and anti-aromatic systems.

Q4 (5-Mark): Describe the ozonolysis of But-2-ene.

Q5 (5-Mark): How does calcium carbide react with water? Write equation.



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: Petrochemicals

Cracking of heavy paraffin fractions produces lower alkenes, key for polymer production.

Application Case: Industrial Solvents

Benzene derivatives act as heavy solvents to manufacture synthetic dyes.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

Aromatic Benzene Ring (C₆H₆) Resonance



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Electrochemical Galvanic Cell System

